Igor Bispo de Moraes Coelho Correia

AI Specialist | Computer Vision & Deep Learning Engineer

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PROFESSIONAL SUMMARY

AI Specialist with 5+ years applying deep learning and computer vision to mission-critical domains — road infrastructure, SAR remote sensing, medical imaging, and automotive safety. Proven track record across the full ML lifecycle: architecture design, training, optimization, and production deployment. MSc in Electrical Engineering (Signal Processing) from the University of Brasília and PhD candidate in Electrical Engineering at the Federal University of Amazonas (UFAM).

TECHNICAL STACK

Deep Learning & CV	PyTorch, TensorFlow, PaddlePaddle, OpenCV, YOLO, YOLOX, RT-DETR, Faster R-CNN, U-Net, Transformers, HuggingFace
Edge AI & Embedded	ONNX Runtime, INT8/FP16 quantization, VeriSilicon ACUITY (NPU), CMake, aarch64 cross-compilation, QEMU, GoogleTest
Languages	Python, C++, C#
Image Processing	SAR imagery, medical imaging (CT/panoramic), embedded video, depth estimation, visual enhancement
MLOps & Infra	Docker, Kubernetes, Apache Airflow, Nagios, Linux on-premise, AWS, GCP
Data	PostgreSQL, SQL Server, Elasticsearch
Tools	CVAT (annotation), Django, Git, GitLab CI/CD, CARLA, FFmpeg

WORK EXPERIENCE

Senior AI Specialist

2025 – present

FPFtech · Manaus, Brazil (hybrid)

Developed computer vision systems for automotive safety: object detection and classification in compliance with international industry regulations, depth estimation, and visual enhancement. Worked across the full CV pipeline — preprocessing, image quality assessment, deep learning inference, and performance monitoring on resource-constrained embedded hardware. Stack: PaddlePaddle/PaddleDetection (PP-YOLOE+), YOLOX, RT-DETR, Depth Anything V2, ONNX/ONNX Runtime, INT8/INT16/FP16 quantization and knowledge distillation, VeriSilicon ACUITY (Vivante VIP9000 NPU), C++17, CMake, aarch64 cross-compilation, QEMU, GoogleTest, GitLab CI/CD, CARLA simulator, FFmpeg and V4L2.

Postgraduate Lecturer — Machine Learning & Deep Learning

2025 – present

IPOS · Remote

Teaching Machine Learning with Python and Deep Learning with Python in the AI for Civil Construction specialization program. Responsible for curriculum design, course delivery, and assessment.

AI Analyst

2022 – 2025

National Transportation Confederation (CNT) · Brasília, Brazil

Managed the full AI model lifecycle — design, training, and deployment — for the CNT Road Survey, Brazil's largest highway assessment program. Built and maintained detection, classification, and analysis systems covering 40+ road variables (pavement defects, signage, viaducts, guardrails). Stack: Python, C++, C#, TensorFlow, PyTorch, OpenCV, PostgreSQL, SQL Server, Apache Airflow.

Postgraduate Researcher

2021 – 2025

University of Brasília (UnB) · Brasília, Brazil

Curupira project: built a deforestation and wildfire detection system for Sentinel-1 SAR imagery using PyTorch and deep convolutional networks. Gigasistêmica project: designed and trained models for osteoporosis and atheroma detection in CT and dental panoramic images (U-Net, semantic segmentation). Supervised annotation pipelines in CVAT.

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Data Scientist *2021 – 2022*

Zerum IT Engineering · Brasília, Brazil

Natural language analysis and fiscal text classification for fraud and anomaly detection. Developed LSTM and Transformer models for time-series data. Production classification API with Django. Stack: Python, TensorFlow, HuggingFace, Elasticsearch.

EDUCATION

PhD in Electrical Engineering (PPGEE) — in progress

Federal University of Amazonas (UFAM) *2026 – present*

MSc in Electrical Engineering — Signal Processing

University of Brasília (UnB) *2022 – 2024*

BSc in Computer Science

University of Brasília (UnB) *2017 – 2021*

PUBLICATIONS

Correia, I. B. M. C. et al. *A Comprehensive Assessment of Optimized YOLO Architectures for Edge and Real-Time Applications*. In: IEEE ICCT-Pacific, 2026. DOI: 10.1109/ICCT-Pacific69083.2026.11518725.

Correia, I. B. M. C. et al. *Importance-Aware Quantization Strategy for Layer-Sensitive YOLO Architectures*. In: IEE ICCT-Pacific, 2026. DOI: 10.1109/ICCT-Pacific69083.2026.11518766.

Correia, I. B. M. C. et al. *Detection and Segmentation of Carotid Atheroma Calcification in Dental Panoramic Radiographs Using a Hybrid Deep Learning Model*. Preprint — Research Square, 2025. Available at: researchsquare.com/article/rs-8078056/v1.

Correia, I. B. M. C.; Farias, M. et al. *Comparative Study of Deforestation Detection in Sentinel-1 Scenes of the Amazon Forest*. In: Proceedings of the XLI Brazilian Symposium on Telecommunications and Signal Processing (SBrT2023). DOI: 10.14209/sbrt.2023.1570923825.

Correia, I. B. M. C. *Deforestation Detection in SAR Images using Deep Neural Networks*. MSc Dissertation — University of Brasília, Brasília, 2024. Available at: repositorio.unb.br/handle/10482/50921.

LANGUAGES

Portuguese Native **English** Advanced **Spanish** Intermediate